

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
First/Second/Third Semester of B. Tech. (All Branches)/ B.Sc. (Physics), BBA/BCA/M.Sc.(IT)

Theory Examination (Regular/Backlog)
April-May 2018

CL142/CL142A Environmental Sciences

Date: 04.05.2018 (Friday)

Time: 01.30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Sections I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q-1. Answer the questions.

[08]

(i) Which of the following is not a layer of atmosphere?

- | | |
|-----------------|------------------|
| (a) Troposphere | (c) Biosphere |
| (b) Mesosphere | (d) Thermosphere |

(ii) Stratosphere contains an important layer of _____ which protects the living species from harmful sun rays.

- | | |
|--------------|--------------|
| (a) Hydrogen | (c) Oxygen |
| (b) Ozone | (d) Nitrogen |

(iii) The most stable ecosystem is _____.

- | | |
|------------|--------------|
| (a) Forest | (c) Ocean |
| (b) Desert | (d) Mountain |

(iv) The animals that feed both on plants and animals are called _____.

- | | |
|-----------------|----------------|
| (a) Omnivores | (c) Carnivores |
| (b) Decomposers | (d) Autotrophs |

(v) The “World Environment Day” is celebrated on _____.

- | | |
|-----------------------------|-------------------------------|
| (a) 15 th July | (c) 5 th June |
| (b) 5 th January | (d) 15 th February |

(vi) _____ is the biggest receptor of pollution on earth.

- | | |
|-----------------|-----------------|
| (a) Lithosphere | (c) Hydrosphere |
| (b) Biosphere | (d) Atmosphere |

(vii) The graphical representation of an organisms position as well as function at each successive trophic level is called _____.

- | | |
|----------------|------------------------|
| (a) Food chain | (c) Ecological pyramid |
| (b) Food web | (d) Biochemical cycle |

(viii) Gunpowder is a mixture of potassium nitrate + _____ + charcoal.

- | | |
|---------------|------------|
| (a) aluminum | (c) lead |
| (b) magnesium | (d) sulfur |

Q-2. Answer any three questions. Each carries five marks. [15]

- (i) Discuss the impacts of technology on environment.
- (ii) Draw the structure of ecosystem and explain it in detail.
- (iii) Explain the types of food chain.
- (iv) Explain Nitrogen cycle in detail.
- (v) Discuss lake ecosystem.

Q-3. Answer any three questions. Each carries four marks. [12]

- (i) Draw a sketch of temperature profile of atmosphere.
- (ii) What is sustainable development? How we can achieve it?
- (iii) Discuss any two ecological pyramids with suitable example.
- (iv) Define ecology. Write the objectives and scope of ecological study.
- (v) Write a short note on Phosphorous Cycle.

SECTION – II**Q-4. Answer the questions. [08]**

- (i) Match the following effect of pollutant on the left hand side with correct pairs of pollutant on the right hand side [04]
- | | |
|-----------------------|-----------------------|
| A. Blue baby diseases | i. Carbon Monoxide |
| B. Minamata disease | ii. Oxides of Sulphur |
| C. Greenhouse effect | iii. Nitrate |
| D. Bronchitis | iv. Nitrous oxide |
| | v. Mercury |
- (ii) Global warming could affect _____ [01]
- | | |
|-------------------------|---------------------|
| (a) Climate | (c) Food production |
| (b) Melting of glaciers | (d) All of them |
- (iii) _____ gives an empirical value to water quality and is a parameter of organic matter present in water. [01]
- | | |
|---------|-----------------|
| (a) BOD | (c) DO |
| (b) COD | (d) All of them |
- (iv) Automobile exhaust consists of _____ [01]
- | | |
|--------------------------------------|---------------------|
| (a) carbon monoxide and nitric oxide | (c) carbon dioxide |
| (b) lead vapour | (d) sulphur dioxide |
- (v) The substances in gaseous or liquid or solid form which pollutes the environment is called _____. [01]
- | | |
|--------------------|---------------|
| (a) eutrophication | (c) soot |
| (b) sag | (d) pollutant |

Q-5. Answer any three questions. Each carries five marks. [15]

- (i) Write a note on acid rain.
- (ii) Explain the ill effects of ozone depletion.
- (iii) Explain the environmental impact of fossil fuels.
- (iv) Write a note on Geothermal and tidal energy.
- (v) Give classification of resources based on regeneration. Quote examples.

Q-6. Answer any three questions. Each carries four marks. [12]

- (i) Name the natural sources of air pollution
- (ii) Discuss the sources of water pollution.
- (iii) Explain the need of conservation of energy resources.
- (iv) Write advantages and disadvantages of conventional energy resources.
- (v) Discuss various sources of noise pollution.
